BRIGMAN

Deep Sea Technology

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Software Requirements Specification (SRS)

Abyss Navigator Suite Project

Brigman Deep Sea Technology

March 6, 2026

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Note on Hyperlinks: Throughout this document, all hyperlinks are displayed in blue and underlined. These links are fully clickable in the PDF viewer and will navigate to the relevant section, requirement, or external resource.

1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the software requirements for the **Abyss Navigator Suite**, an unmanned deep-sea submersible fleet control system for Triton Deepwater. It is intended to be the baseline agreement for design, implementation, verification, and change control.

1.2 Document Conventions

Template source. This SRS is structured using the *Software Requirements Specification* template by **Karl E. Wiegers (1999)** (provided in `References/srs_template-ieee.pdf`), adapted to incorporate the course rubric requirements (codification, iRTM, and change mitigation).

Normative language. All requirements use **shall** to indicate mandatory behavior.

Codification (mandatory). The following ID schemes are used throughout:

- **Software requirements:** `<DOMAIN>-<FEATURE>-<NNN>` (e.g., `SAF-EMG-001`).
- **Technology requirements:** `TECH-<AREA>-<NNN>` (e.g., `TECH-PLAT-001`).
- **Quality model features:** `QUAL-<AREA>-<NNN>` (e.g., `QUAL-SAF-001`).
- **iRTM entries:** `RTM-<NNN>` (e.g., `RTM-001`).
- **Change mitigation controls:** `CHG-<AREA>-<NNN>` (e.g., `CHG-CONT-001`, `CHG-OISS-001`, `CHG-HRCC-001`).

Priority levels. Priority is recorded as **Essential**, **Desirable**, or **Acceptable** to align with the Project Specification.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for Triton Deepwater stakeholders, the Brigman development and verification teams, project management, and independent auditors. Readers should begin with Sections 1–3 (scope, constraints, codification) before using Section 4 (requirements) and Sections 7–8 (traceability and change mitigation).

1.4 Product Scope

The Abyss Navigator Suite enables unmanned missions executed autonomously, remotely supervised from a surface or mothership environment, with controlled intervention, safe emergency handling, and operational traceability.

2 Overall Description

2.1 Product Functions

At a high level, the system provides:

- Mission planning and autonomous execution for unmanned deep-sea missions.
- Remote monitoring, operator alerting, and controlled remote intervention.
- Fleet-level supervision for multiple submersibles concurrently.

- Emergency detection, safe-state transition, and recovery coordination.
- Secure operational data management, logging, and post-mission traceability.

2.2 User Classes and Characteristics

- **Mission Supervisor:** monitors mission progress and authorizes intervention.
- **Fleet Operator:** manages multi-vehicle assignments and scheduling.
- **Safety & Compliance Officer:** reviews anomalies, audits logs, and ensures governance obligations are met.
- **System Administrator:** manages credentials, configuration, and system health.
- **Data Analyst:** exports mission data for downstream analysis.

2.3 Operating Environment (Latest Tech)

The operating environment is characterized by remote supervision from a mothership environment and constrained connectivity to submersibles. The vendor-selected software platform is described in Section 5. Key operating constraints include:

- Underwater communications may be intermittent and low-bandwidth.
- Extreme pressure and low visibility increase reliance on sensor-driven autonomy.
- Operational data is safety- and audit-relevant and must be protected.

2.4 Design and Implementation Constraints

The following constraints apply:

- **Remote operations context:** mission supervision and control are performed remotely from surface/mothership environments.
- **Communications constraints:** the system must operate under constrained bandwidth/latency and intermittent connectivity.
- **Security and governance:** operational data protection and auditability are required.
- **Safety-first policy:** failures and abnormal conditions must bias toward safe-state behavior.

2.5 User Documentation

The following documentation shall be delivered with the software:

- Operator manual (mission planning, supervision, interventions, safety procedures).
- System administration guide (credentials, configuration, deployment).
- Interface/API guide for integrating Triton-provided sensors and external systems.

2.6 Assumptions and Dependencies

- Triton provides mission sensors and their interface specifications for integration.
- Missions are supervised by trained and authorized personnel.
- Regulatory obligations apply to maritime safety and data governance.

3 Codification

3.1 Codification Rules

Requirement IDs are stable across revisions. The numeric suffix increments only when a new requirement is added; edited requirements retain their IDs to preserve traceability in the iRTM.

3.2 Requirement Attributes and Quality Tagging

Each requirement in Section 4 is recorded with:

- **Source:** originating Project Specification clause(s) (e.g., `BO-L1-4`, `IT-L0-C3`).
- **Priority:** Essential / Desirable / Acceptable.
- **Quality tags:** one or more `QUAL-*` IDs from Section 6.
- **Verification:** inspection, analysis, test, or demonstration.
- **Use Case link:** reference to the primary use case(s) in Figure 1 (numbered `UC01`, `UC02`, ...) that realize or exercise the requirement.

4 Specific Requirements

4.1 Navigation & Mission Autonomy

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
NAV-PLAN-001	Essential	BO-L1-1	QUAL-OPS-001, QUAL-TRC-001	The system shall allow an authorized operator to create and version a mission plan containing an ordered set of waypoints (latitude, longitude, depth) and mission phases.	Demo: Create/edit/version plan; inspect stored plan versions.	UC01 (Mission planning)
NAV-EXEC-001	Essential	BO-L1-1, BO-L0-C1	QUAL-RES-001, QUAL-SAF-001	The system shall execute a validated mission plan autonomously without continuous human control for a full mission cycle in a representative simulation.	Test: End-to-end mission simulation with no operator commands.	UC02 (Autonomous mission execution)
NAV-WPT-001	Essential	BO-L1-1	QUAL-OPS-001	The system shall achieve **99% waypoint arrival success** in which each reached waypoint is within $\leq$5 m horizontal and $\leq$2 m vertical error under representative sea-trial conditions.	Analysis: Compare logged track vs. waypoints across trials.	UC02 (Waypoint navigation)
NAV-FLEET-001	Desirable	BO-L1-3, IT-L1-3	QUAL-OPS-001, QUAL-RES-001	The system shall support concurrent supervision of at least **3** submersibles, including assignment of mission plans and real-time status visibility per vehicle.	Test: Simulated 3-vehicle session; verify assignment and visibility.	UC03 (Supervise multiple vehicles)

Table 2: Navigation & Mission Autonomy Requirements

4.2 Communications & Remote Operations

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
COM-MON-001	Essential	BO-L1-2, IT-L1-1	QUAL-OPS-001	The system shall provide remote mission and vehicle status telemetry updates at least once every **30 seconds** when connectivity is available.	Test: Measure telemetry update intervals under nominal link.	UC04 (Monitor telemetry)
COM-INT-001	Essential	IT-L1-2, BO-L1-2	QUAL-OPS-001, QUAL-SAF-001	The system shall allow an authorized operator to issue remote intervention commands (pause, resume, safe-state) and receive acknowledgement within $\leq$ **60 seconds** when the link latency is $\leq$30 seconds.	Test: Command/ack timing under simulated 30s latency.	UC05 (Remote intervention)
COM-DEG-001	Essential	IT-L0-C2, IT-L1-6	QUAL-RES-001, QUAL-SAF-001	The system shall support safe operation under constrained communications, including operation at $\geq$ **1 kbps** bandwidth, $\leq$ **30 s** one-way latency, and intermittent connectivity, and shall enter safe-state if connectivity is lost continuously for $>$ **10 minutes**.	Test: Fault-inject bandwidth, latency, and outage; verify safe-state trigger.	UC06 (Handle degraded comms)

Table 3: Communications & Remote Operations Requirements

4.3 Safety & Recovery

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
SAF-EMG-001	Essential	BO-L1-4, BO-L2-4.1	QUAL-SAF-001, QUAL-OPS-001	The system shall detect pressure deviations $> \pm 5\%$ from nominal setpoint within **5 seconds**.	Test: Inject simulated pressure spike; measure detection latency.	UC07 (Detect anomalies)
SAF-EMG-002	Essential	BO-L1-4, BO-L2-4.1	QUAL-SAF-001, QUAL-OPS-001	The system shall detect temperature deviations $> \pm 10\%$ from nominal within **10 seconds**.	Test: Thermal chamber simulation or sensor-in-the-loop test.	UC07 (Detect anomalies)
SAF-EMG-003	Essential	BO-L1-4, BO-L2-4.1	QUAL-SAF-001, QUAL-OPS-001	The system shall detect power supply deviations $> \pm 15\%$ from nominal within **10 seconds**.	Test: Power supply fault injection; measure detection latency.	UC07 (Detect anomalies)

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Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
SAF-STATE-001	Essential	BO-L2-4.2	QUAL-SAF-001	Upon detection of any abnormal condition, the system shall transition the submersible to a **safe operational state** defined as neutral buoyancy, minimal power consumption, and active communication beacon.	Integration test: End-to-end emergency scenario.	UC08 (Enter safe state)
SYS-CONT-001	Essential	BO-L1-5	QUAL-RES-001, QUAL-SAF-001	When non-critical degradation occurs, the system shall continue the mission in a degraded mode while alerting the operator and logging the degradation event.	Test: Inject non-critical fault; confirm degraded continuation and alert/log.	UC09 (Continue in degraded mode)
SYS-TERM-001	Essential	BO-L1-5	QUAL-SAF-001, QUAL-TRC-001	When termination is required, the system shall terminate the mission in an orderly manner by completing the current operation when safe, logging the shutdown reason, and entering safe-state.	Test: Inject critical fault; verify ordered termination sequence.	UC10 (Terminate mission)
SYS-REC-001	Essential	BO-L1-5	QUAL-SAF-001, QUAL-OPS-001	After entering safe-state, the system shall support recovery coordination by continuously broadcasting a communication beacon and exposing last-known position and status to operators until recovery is confirmed.	Demo/Test: Simulate safe-state; verify beacon and status visibility.	UC11 (Coordinate recovery)

Table 4: Safety & Recovery Requirements

4.4 Data Logging & Accountability

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
DATA-LOG-001	Essential	BO-L1-6, IT-L1-5	QUAL-TRC-001	The system shall log mission events and operator actions with timestamp (UTC), vehicle ID, and operator ID attribution.	Inspection/Test: Review log records for required fields.	UC12 (Record operations)
DATA-LOG-002	Essential	IT-L1-5, IT-L0-C3	QUAL-SEC-001, QUAL-TRC-001	The system shall store operational logs in an append-only form and retain them for at least **5 years** to support audit and investigation.	Inspection: Verify storage configuration and retention policy.	UC12 (Record operations)
DATA-LOG-003	Essential	BO-L2-6.2, IT-L2-5.1	QUAL-OPS-001, QUAL-TRC-001	The system shall allow authorized retrieval and export of mission records for a $\leq$24-hour mission into a structured format (CSV and JSON) within $\leq$ **2 minutes**.	Test: Export timed retrieval under representative data volume.	UC13 (Re-trieve/export records)

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Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
DATA-LOG-004	Desirable	BO-L1-7, BO-L2-6.2	QUAL-OPS-001	The system shall generate a post-mission summary report (mission timeline, anomalies, interventions) within $\leq$ **10 minutes** after mission completion.	Test: Complete mission; measure report generation time and contents.	UC14 (Post-mission summary)

Table 5: Data Logging & Accountability Requirements

4.5 External Interface Requirements

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
INT-SENS-001	Essential	IT-L0-A1	QUAL-RES-001	The system shall integrate customer-supplied mission sensors using **ROS2 middleware** and **DDS** transport, and shall ingest sensor messages in a documented **JSON** schema defined in the Interface/API guide.	Test: Sensor-in-the-loop integration against the published schema.	UC15 (Sensor integration)

Table 6: External Interface Requirements

4.6 Security & Governance

Req ID	Priority	Source	Quality Tags	Requirement	Verification	Use Case Link
SEC-ENC-001	Essential	IT-L0-C3, IT-L1-4	QUAL-SEC-001	The system shall protect operational data using **AES-256** encryption for data at rest and for encrypted payloads in transit.	Inspection/Test: Verify crypto configuration and encrypted traffic capture.	UC16 (Protect data)
SEC-AUTH-001	Essential	IT-L0-C3, IT-L1-4	QUAL-SEC-001	The system shall enforce mutual authentication for system-to-system communications using **X.509 certificates**.	Test: Reject invalid/expired certificates; accept valid certificates only.	UC16 (Mutual authentication)
SEC-AUD-001	Essential	IT-L0-C3, BO-L0-C4	QUAL-SEC-001, QUAL-TRC-001	The system shall maintain audit logs for access to operational data and for administrative configuration changes, including actor identity and timestamps.	Inspection/Test: Perform actions; confirm corresponding audit records.	UC17 (Audit access)

Table 7: Security & Governance Requirements

5 Technology Requirements (Latest Tech)

The following technology requirements define the intended operating platform, data store, and interface patterns selected by Brigman Deep Sea Technology to satisfy the Project Specification constraints.

Tech ID	Area	Specification
TECH-PLAT-001	Platform	Vehicle-side software components shall run on a Linux-based OS with ROS2 middleware support; mothership-side services shall support deployment on standard workstation/server environments.
TECH-DB-001	Database	The system shall store operational and mission records in a structured database that supports retention policies and encrypted storage (e.g., PostgreSQL or equivalent).
TECH-NET-001	Networking	The system shall support message transport compatible with constrained links (DDS where available) and shall support store-and-forward buffering during outages.
TECH-API-001	APIs	External and internal service interfaces shall use versioned JSON schemas and provide a documented API contract for integration and export.

Table 8: Technology requirements for operating environment and platform

6 Quality Development Model

Brigman Deep Sea Technology adopts BDST Quality Model v1.1, which is explicitly aligned with McCall's Software Quality Model (1977) across the three perspectives: Product Operation, Product Revision, and Product Transition.

6.1 6.1 Alignment with McCall Quality Model

McCall Perspective	McCall Feature	Covered by BDST Quality ID
Product Operation	Reliability	QUAL-SAF-001, QUAL-RES-001
Product Operation	Integrity	QUAL-SEC-001
Product Operation	Usability	QUAL-OPS-001
Product Operation	Correctness	NAV-WPT-001, SAF-EMG-001..003
Product Operation	Efficiency	COM-MON-001 (timing constraints)
Product Revision	Maintainability	TECH-API-001, TECH-DB-001
Product Revision	Testability	Verification column (Section 4)
Product Revision	Flexibility	SYS-CONT-001 (degraded mode handling)
Product Transition	Interoperability	INT-SENS-001
Product Transition	Portability	TECH-PLAT-001
Product Transition	Reusability	Modular API schema (TECH-API-001)

Table 9: Mapping of BDST Quality Model to McCall Quality Model

6.2 6.2 BDST Quality Features

- QUAL-SAF-001 (Safety Integrity): Measures detection latency, safe-state transition correctness, and recovery robustness under fault injection.
- QUAL-SEC-001 (Security & Governance): Measures encryption coverage, authentication enforcement, and audit completeness.

- QUAL-RES-001 (Resilience): Measures safe operation under degraded communications and abnormal conditions.
- QUAL-TRC-001 (Traceability): Measures reconstruction capability of mission timeline with actor attribution.
- QUAL-OPS-001 (Operability & Performance): Measures telemetry frequency, detection timing, intervention response, and report generation performance.

7 Integrated Requirements Traceability Matrix (iRTM)

The iRTM traces each Project Specification clause to the corresponding SRS requirement(s) or SRS section(s), using minimized paraphrases to preserve CRaM meaning while remaining concise.

7.1 Traceability Governance Principles

The iRTM establishes the initial chain-of-custody between the Project Specification (PS) and this SRS. It is governed by the following principles:

- Every PS clause shall map to at least one SRS requirement or SRS section.
- Every SRS requirement shall have a trace origin (PS clause or Vendor Provision).
- Requirement identifiers remain stable across revisions to preserve trace integrity.
- Outline numbering shall not be used for trace identification.
- The iRTM is an inceptive trace artifact and shall be extended during Design, Implementation, and Verification phases.

7.2 Trace Type Legend

Trace links in this iRTM are classified as:

- **DT (Direct Trace)**: PS clause directly generates one or more SRS requirement IDs.
- **AT (Assumption Trace)**: PS clause is captured as an assumption/dependency in the SRS, traced to the relevant SRS section.
- **VP (Vendor Provision)**: Requirement introduced by the vendor as part of Specifying method (e.g., technology provisions, quality model provisions, change protocol).

iRTM ID	PS Ref	Trace Type	PS Clause (min.)	SRS Element	SRS Clause / Coverage (min.)
RTM-001	BO-L0-C1	DT	Unmanned operations at extreme depths.	NAV-EXEC-001	Autonomous execution without continuous human control.
RTM-002	BO-L0-C2	DT	Safety-first: eliminate human exposure.	SAF-STATE-001; SYS-TERM-001	Safe-state bias and orderly termination under risk.
RTM-003	BO-L0-C3	DT	Extreme pressure/low visibility/limited comm.	COM-DEG-001; SAF-EMG-001	Constrained-link behavior; abnormal detection.

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iRTM ID	PS Ref	Trace Type	PS Clause (min.)	SRS Element	SRS Clause / Coverage (min.)
RTM-004	BO-L0-C4	DT	Comply with maritime safety/data governance.	SEC-AUD-001; DATA-LOG-002	Auditable access and retained records.
RTM-005	BO-L0-A1	AT	Missions supervised by trained personnel.	Section 2.6	Operational assumption recorded (software supports supervision).
RTM-006	BO-L1-1	DT	Autonomous mission execution required.	NAV-PLAN-001; NAV-EXEC-001; NAV-WPT-001	Plan, autonomous execution, and waypoint performance.
RTM-007	BO-L1-2	DT	Remote mission oversight required.	COM-MON-001; COM-INT-001	Telemetry updates and remote interventions.
RTM-008	BO-L1-3	DT	Coordinated multi-submersible operation.	NAV-FLEET-001	Concurrent supervision and mission assignment.
RTM-009	BO-L1-4	DT	Handle abnormal/emergency conditions.	SAF-EMG-001..003; SAF-STATE-001	Detection thresholds and safe-state transition.
RTM-010	BO-L1-5	DT	Continue safely or terminate orderly.	SYS-CONT-001; SYS-TERM-001; SYS-REC-001	Degraded continuation, termination, recovery coordination.
RTM-011	BO-L1-6	DT	Accountability for mission activities/decisions.	DATA-LOG-001; SEC-AUD-001	Operator-attributed logging and auditability.
RTM-012	BO-L1-7	DT	Outputs usable for ops and analytics.	DATA-LOG-003; DATA-LOG-004	Exportable records and summary reporting.
RTM-013	BO-L2-4.1	DT	Abnormal conditions identifiable.	SAF-EMG-001..003	Defined abnormal thresholds and detection timing.
RTM-014	BO-L2-4.2	DT	Transition to safe state on critical.	SAF-STATE-001	Defined safe operational state behavior.
RTM-015	BO-L2-6.1	DT	Activities/decisions traceable.	DATA-LOG-001; SEC-AUD-001	Trace logs and audit logs with actor identities.
RTM-016	BO-L2-6.2	DT	Records support review/investigation.	DATA-LOG-002..004	Retention, export, and post-mission reporting.
RTM-017	IT-L0-C1	DT	Remote supervision/control context.	COM-MON-001; COM-INT-001	Remote monitoring and intervention functions.
RTM-018	IT-L0-C2	DT	Constrained band-width/latency/intermittent.	COM-DEG-001	Explicit bandwidth, latency, and outage thresholds.
RTM-019	IT-L0-C3	DT	Protect operational data per obligations.	SEC-ENC-001; SEC-AUTH-001; SEC-AUD-001	Encryption, authentication, audit.

Continued on next page

iRTM ID	PS Ref	Trace Type	PS Clause (min.)	SRS Element	SRS Clause / Coverage (min.)
RTM-020	IT-L0-A1	DT	Customer provides sensors for integration.	INT-SENS-001	ROS2/DDS/JSON-based sensor integration.
RTM-021	IT-L1-1	DT	Remote monitoring capability required.	COM-MON-001	Telemetry update rate requirement.
RTM-022	IT-L1-2	DT	Remote intervention capability required.	COM-INT-001	Remote command/ack requirement.
RTM-023	IT-L1-3	DT	Multi-vehicle supervision required.	NAV-FLEET-001	3-vehicle concurrent supervision.
RTM-024	IT-L1-4	DT	Protect confidentiality/integrity of data.	SEC-ENC-001; SEC-AUTH-001	Encryption and mutual authentication.
RTM-025	IT-L1-5	DT	Logging mission events and operator actions.	DATA-LOG-001..004	Logging, retention, export, reporting.
RTM-026	IT-L1-6	DT	Safe operation under degraded comm.	COM-DEG-001; SYS-CONT-001	Degraded comm handling and mission continuity.
RTM-027	IT-L2-5.1	DT	Logs sufficient for review/accountability.	DATA-LOG-003; DATA-LOG-004	Exportable records and review-ready reports.
RTM-028	Vendor Provision	VP	Vendor-defined technology stack and integration standards.	TECH-PLAT-001; TECH-DB-001; TECH-NET-001; TECH-API-001	Technology requirements introduced under the Specifying method.
RTM-029	Vendor Provision	VP	Vendor-defined quality development model.	QUAL-SAF-001; QUAL-SEC-001; QUAL-RES-001; QUAL-TRC-001; QUAL-OPS-001	Quality attributes structured under the McCall-aligned development model.
RTM-030	Vendor Provision	VP	Vendor-defined change-mitigation controls.	CHG-CONT-001..003; CHG-OISS-001..002; CHG-HRCC-001..004	Change governance introduced under Section 8.

Table 10: iRTM: Project Specification (PS) to SRS traceability

7.3 SDLC Trace Extension

The iRTM is an inceptive trace artifact. During subsequent SDLC phases, the trace matrix shall be extended with references to:

- Design Module Identifier (architecture/subsystem mapping)
- Test Case Identifier (verification and acceptance test cases)
- Repository Artifact Reference (implementation modules/commits)

No requirement shall progress to implementation without corresponding design and verification references.

8 Change Mitigation Protocol

This protocol implements all three change mitigation strategies defined in the Specifying methodology: anticipatory contingency controls, open issue controls, and high-risk change controls. All change mitigation items are codified under the scheme `CHG-<AREA>-<NNN>` as declared in Section 3.

8.1 8.1 Contingency Controls

All contingency-related change controls are codified under `CHG-CONT-<NNN>`. The following contingency controls are pre-approved and may be activated without contract renegotiation if triggering conditions occur.

- **CHG-CONT-001:** If regulatory obligations change during project execution, the baseline shall be reviewed and updated through formal impact analysis before implementation.
  - Impact assessment required.
  - Must be trace-linked in iRTM.
  - Must update affected requirement IDs.
- **CHG-CONT-002:** If customer-provided sensor interfaces deviate from agreed specifications, integration shall be suspended until compatibility validation is completed.
  - Impact assessment required.
  - Must be trace-linked in iRTM.
  - Must update affected requirement IDs (`INT-SENS-001`).
- **CHG-CONT-003:** If communications constraints exceed defined thresholds, degraded-mode operation shall be activated and logged.
  - Impact assessment required.
  - Must be trace-linked in iRTM.
  - Must update affected requirement IDs (`COM-DEG-001`, `SYS-CONT-001`).

8.2 8.2 Open Issue Controls

Open issues shall be codified under `CHG-OISS-<NNN>`. Open issues are requirements not fully resolved at SRS baseline and subject to structured negotiation.

Issue ID	Description	Responsible Party	Kill Date
CHG-OISS-001	Final confirmation of regulatory audit retention duration is pending.	Vendor	30 days post-baseline
CHG-OISS-002	Final bandwidth constraint values subject to field validation.	Customer	45 days post-baseline

Table 11: Open Issue Controls Register

Open issues:

- Must not invalidate approved requirements.
- Must be resolved before baseline freeze.
- Must be tracked until closure.

8.3 8.3 High-Risk Change Controls

High-risk changes shall be codified under `CHG-HRCC-<NNN>`. High-risk changes include architectural modifications, safety-state logic changes, security mechanism replacement, and autonomy execution logic changes.

- **CHG-HRCC-001:** Any modification to safe-state transition logic requires dual review and regression validation.
- **CHG-HRCC-002:** Replacement of encryption mechanism requires security impact assessment and re-validation.
- **CHG-HRCC-003:** Changes affecting autonomous navigation behavior require simulation validation prior to deployment.
- **CHG-HRCC-004:** Changes affecting trace logging or auditability require trace matrix update before approval.

All high-risk changes require:

1. Formal impact analysis (safety, schedule, cost).
2. Stakeholder approval (Customer Safety Approver and Vendor Authenticator).
3. Trace update in iRTM.
4. Re-validation per Section 6.

9 Addenda

Addendum ID	Statement
ADD-001	**Project direction baseline:** This SRS assumes the customer direction change to a fully unmanned, autonomous-first operating model is in force for all missions.
ADD-002	**Cost redaction note:** Any cost figures referenced from the Project Specification are treated as indicative and may be redacted in released versions to avoid biasing proposals.

Table 12: Addenda (purpose-appropriate notes adopted from the Project Specification)

10 Requirements Modeling

The following models are included to support stakeholder understanding and to maintain traceability between requirements and operational context.

- **Use Case Diagram** (Figure 1) – depicts the consolidated actor–system interactions, covering all major use cases referenced throughout this SRS.
- **Data Dictionary** (Appendix A) – defines all data entities, their attributes, data types, constraints, and descriptions that underpin the system's data model. Each entity is cross-referenced to the relevant use case flows in Appendix B and C.

Additional Project Specification models (e.g., organization chart and roles/responsibilities) are adopted by reference from the Project Specification appendices to avoid divergence across artifacts.

11 Sign-off

11.1 Vendor Authentication

Name	Employee ID	Signature
Muhammad Ridwan Putra Jasni (Authenticator)	BDST-RE-BA-001	

Table 13: Vendor authenticator sign-off (signature provided)

11.2 Designated Approvers (No Signatures Required)

Organization	Role	Name (Designated)
Customer (Triton Deepwater)	Project Leader / Requirements Approver	Sarah Whitman
Customer (Triton Deepwater)	Safety & Compliance Approver	(TBD)
Vendor (Brigman)	Engineering Delivery Lead	(TBD)
Vendor (Brigman)	Verification & Validation Lead	(TBD)

Table 14: Designated approvers (placeholders where not yet nominated)

Repository Information

PROJECT REPOSITORY

https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113

SRS, models, and traceability artifacts maintained under version control

References

- Triton Deepwater, *Project Specification: The Abyss Navigator Suite*, ICT2113, 2026.
- Brigman Deep Sea Technology, *Software Requirements Analysis Report (BCIC Methodology)*, 2026.
- Kevin Anthony Jones, *Labtorial Guide for Class Project: Reqrs now & near future*, Dec 2025.
- Kevin Anthony Jones, *Guide Rubric for Assessment of Deliverables*, Jan 2026.
- Karl E. Wiegers, *Software Requirements Specification Template*, 1999.

Glossary

- **BCIC:** Breakdown, Clarify, Interpret, Categorize.
- **CRaM:** Correct, Relevant, and Measurable requirements organization principle.
- **iRTM:** Integrated Requirements Traceability Matrix.
- **Safe-state:** neutral buoyancy, minimal power, beacon active.

To Be Determined (TBD) List

- Customer nomination of Safety & Compliance approver name.
- Final selection of database product/version for TECH-DB-001 (if different from PostgreSQL-equivalent).
- Final JSON schemas for INT-SENS-001 (to be published with the Interface/API guide).

Figures

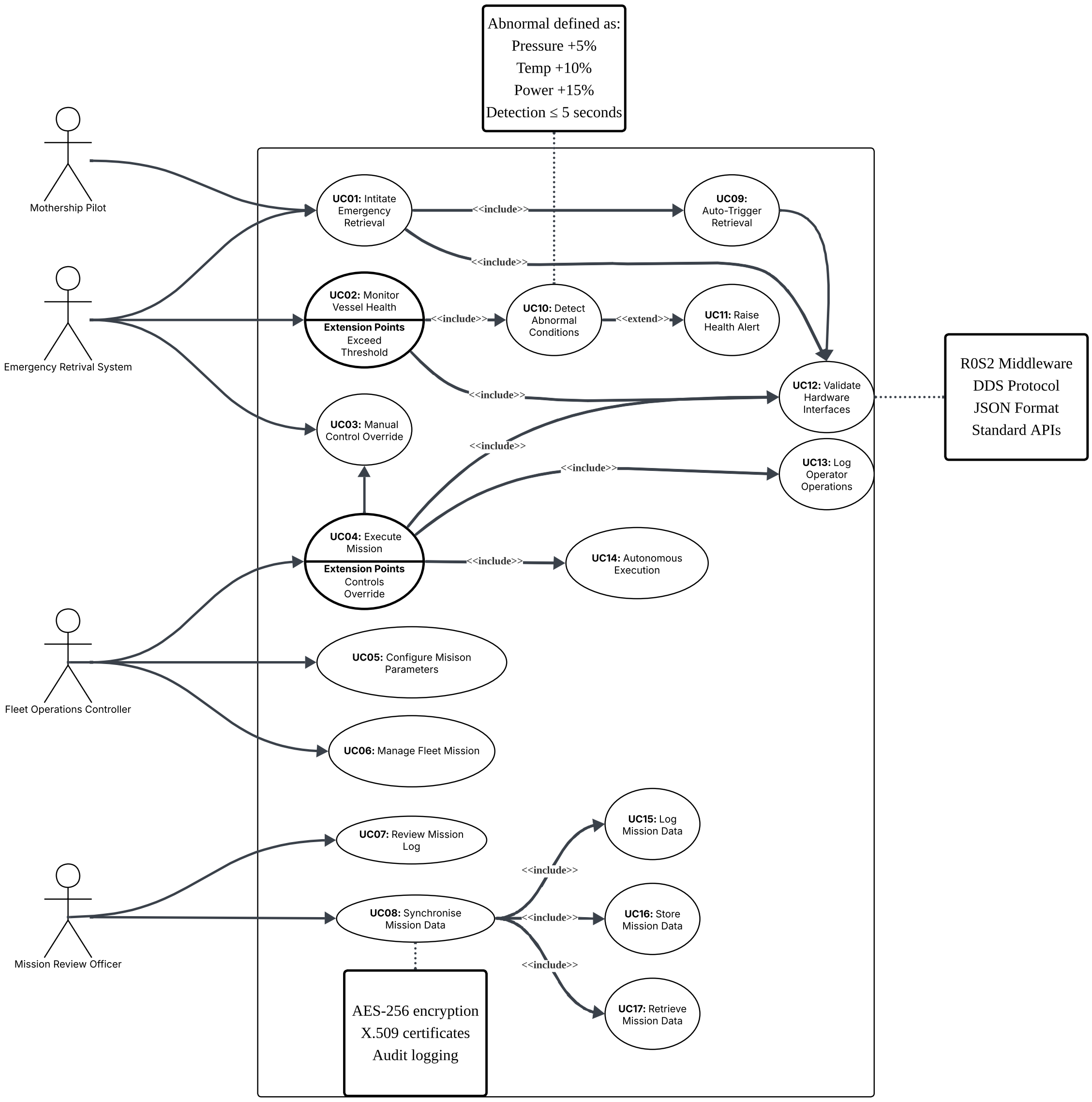

Figure 1: Use Case Diagram

Instructions

The diagram illustrates the primary actors (surface controller, mothership crew, autonomous submersible) and their interactions with major system functions such as mission planning, launch/recovery, sensor telemetry, fault handling, and data logging. Follow each actor's swim lane to see how the mission progresses from preparation to post-mission data review; the colored connectors highlight safety-critical flows (emergency surfacing, manual override) that must remain available throughout operations.

APPENDICES

Appendix A: Data Dictionary

This section defines the data entities used by the system and the attributes that describe each entity.

Entity: Operator				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
OperatorID	VARCHAR	20	PK, NOT NULL, UNIQUE	Unique operator identifier used in UI (e.g., OP-2847)
Username	VARCHAR	50	NOT NULL, UNIQUE	Login username / operator ID entry
FullName	VARCHAR	100	NOT NULL	Display name shown in admin list
Role	ENUM	-	'Fleet Operator', 'Mission Supervisor', 'Safety & Compliance Officer', 'Data Analyst', 'System Administrator'	Selected user role
AccountStatus	ENUM	-	'Active', 'Inactive', 'Locked', 'Disabled'	Account status shown in admin UI
TwoFactorEnabled	BOOLEAN	-	NOT NULL, DEFAULT FALSE	Whether 2FA is enabled for this user
LastLoginUTC	TIMESTAMP	-	NULL	Last successful login time (UTC)
CreatedAtUTC	TIMESTAMP	-	NOT NULL	Record creation time (UTC)

Table A-1: Data dictionary for **Operator** entity

Entity: LoginSession				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
SessionID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique session identifier
OperatorID	VARCHAR	20	FK → Operator.OperatorID, NOT NULL	Logged-in user
LoginUTC	TIMESTAMP	-	NOT NULL	Login timestamp (UTC)
LogoutUTC	TIMESTAMP	-	NULL	Logout timestamp (UTC)
TwoFactorVerified	BOOLEAN	-	NOT NULL	Whether 2FA verification succeeded
SessionStatus	ENUM	-	'Active', 'Expired', 'Terminated'	Session state
IPAddress	VARCHAR	45	NULL	Optional login source IP address

Table A-2: Data dictionary for **LoginSession** entity

Entity: Vehicle				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
VehicleID	VARCHAR	20	PK, NOT NULL, UNIQUE	Vehicle identifier shown in UI (e.g., ASV-01)
DisplayName	VARCHAR	100	NOT NULL	Vehicle name shown (e.g., Deep Explorer)
OperationalStatus	ENUM	-	'Active', 'Paused', 'Safe-State'	Status shown on vehicle card
CurrentDepth_m	DECIMAL	(8,2)	NULL	Current depth (meters) shown on vehicle card
CurrentPower_pct	DECIMAL	(5,2)	NULL	Current power (percent) shown on vehicle card
CurrentPressure_bar	DECIMAL	(7,2)	NULL	Current pressure (bar) shown on vehicle card
CurrentTemp_c	DECIMAL	(5,2)	NULL	Current temperature (Celsius) shown on vehicle card
LastKnownLat	DECIMAL	(10,6)	NULL	Last known latitude
LastKnownLon	DECIMAL	(10,6)	NULL	Last known longitude
LastUpdateUTC	TIMESTAMP	-	NULL	Last telemetry update time (UTC)

Table A-3: Data dictionary for **Vehicle** entity

Entity: MissionPlan				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
MissionPlanID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique mission plan identifier
MissionName	VARCHAR	120	NOT NULL	Mission name shown in UI (e.g., Hydrothermal Survey 12B)
Version	VARCHAR	10	NOT NULL	Mission plan version (e.g., 1.3)
TargetVehicleID	VARCHAR	20	FK → Vehicle.VehicleID, NOT NULL	Selected target vehicle
Estimated DurationHours	DECIMAL	(4,1)	NULL	Estimated duration (hours) shown in UI
Status	ENUM	-	'Draft', 'Deployed', 'Archived'	Mission plan status shown in summary
CreatedBy	VARCHAR	20	FK → Operator.OperatorID, NOT NULL	Creator of the plan
CreatedAtUTC	TIMESTAMP	-	NOT NULL	Creation time (UTC)
UpdatedAtUTC	TIMESTAMP	-	NULL	Last update time (UTC)

Table A-4: Data dictionary for **MissionPlan** entity

Entity: Waypoint				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
WaypointID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique waypoint identifier
MissionPlanID	VARCHAR	36	FK → MissionPlan. MissionPlanID, NOT NULL	Mission plan this waypoint belongs to
SequenceNo	INTEGER	5	NOT NULL	Waypoint order (supports reordering in UI)
Latitude	DECIMAL	(10,6)	NOT NULL	Waypoint latitude
Longitude	DECIMAL	(10,6)	NOT NULL	Waypoint longitude
Depth_m	DECIMAL	(8,2)	NOT NULL	Target depth at this waypoint (meters)
Phase	ENUM	-	'Transit', 'Survey', 'Sample'	Operational phase selected in UI

Table A-5: Data dictionary for **Waypoint** entity

Entity: MissionRun				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
MissionID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique mission execution identifier
MissionPlanID	VARCHAR	36	FK → MissionPlan. MissionPlanID, NOT NULL	Mission plan deployed
VehicleID	VARCHAR	20	FK → Vehicle.VehicleID, NOT NULL	Vehicle executing mission
StartUTC	TIMESTAMP	-	NOT NULL	Mission start time (UTC)
EndUTC	TIMESTAMP	-	NULL	Mission end time (UTC)
State	ENUM	-	'Running', 'Completed', 'Paused', 'Safe-State', 'Terminated'	Mission execution state
WaypointSuccess_ pct	DECIMAL	(5,2)	NULL	Waypoint success rate (percent)

Table A-6: Data dictionary for **MissionRun** entity

Entity: TelemetrySnapshot				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
TelemetryID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique telemetry record
MissionID	VARCHAR	36	FK → MissionRun.MissionID, NOT NULL	Mission reference
VehicleID	VARCHAR	20	FK → Vehicle.VehicleID, NOT NULL	Vehicle reference
TimestampUTC	TIMESTAMP	-	NOT NULL	Time telemetry recorded (UTC)
Depth_m	DECIMAL	(8,2)	NOT NULL	Depth (meters) shown on vehicle card
Power_pct	DECIMAL	(5,2)	NOT NULL	Power (percent) shown on vehicle card
Pressure_bar	DECIMAL	(7,2)	NOT NULL	Pressure (bar) shown on vehicle card
Temp_c	DECIMAL	(5,2)	NOT NULL	Temperature (Celsius) shown on vehicle card
Latitude	DECIMAL	(10,6)	NOT NULL	Latitude shown on vehicle card and map
Longitude	DECIMAL	(10,6)	NOT NULL	Longitude shown on vehicle card and map
PayloadJSON	JSON	-	NULL	Optional raw payload for extensibility

Table A-7: Data dictionary for **TelemetrySnapshot** entity

Entity: CommsLinkSnapshot				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
LinkSnapID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique communications snapshot
MissionID	VARCHAR	36	FK → MissionRun.MissionID, NOT NULL	Mission reference
TimestampUTC	TIMESTAMP	-	NOT NULL	Snapshot time (UTC)
LinkType	ENUM	-	'Satellite', 'Acoustic', 'RF'	Communication link type used
ConnectionStatus	ENUM	-	'Connected', 'Degraded', 'Disconnected'	Link connection status
LinkQuality_pct	DECIMAL	(5,2)	NOT NULL	Link quality (percent) shown in UI
Bandwidth_mbps	DECIMAL	(6,2)	NOT NULL	Bandwidth (Mbps) shown in UI
Latency_ms	INTEGER	6	NOT NULL	Latency (ms) shown in UI

Table A-8: Data dictionary for **CommsLinkSnapshot** entity

Entity: InterventionCommand				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
CommandID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique command identifier
MissionID	VARCHAR	36	FK → MissionRun.MissionID, NOT NULL	Related mission
TargetVehicleID	VARCHAR	20	FK → Vehicle.VehicleID, NOT NULL	Vehicle receiving the command
IssuedBy	VARCHAR	20	FK → Operator.OperatorID, NOT NULL	Operator who issued the command
CommandType	ENUM	-	'Pause', 'Resume', 'Safe-State'	Command type from the dashboard controls
IssuedAtUTC	TIMESTAMP	-	NOT NULL	Command issued time (UTC)
CommandStatus	ENUM	-	'Sent', 'Acknowledged', 'Failed'	Command delivery outcome
AckReceivedAtUTC	TIMESTAMP	-	NULL	Acknowledgement received time (UTC)
FailureReason	VARCHAR	255	NULL	Reason if command failed

Table A-9: Data dictionary for **InterventionCommand** entity

Entity: BaselineProfile				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
BaselineID	VARCHAR	36	PK, NOT NULL, UNIQUE	Baseline profile identifier
VehicleID	VARCHAR	20	FK → Vehicle.VehicleID, NOT NULL	Vehicle baseline applies to
MetricType	ENUM	-	'Power', 'Pressure', 'Temp'	Metric being compared against baseline
BaselineValue	DECIMAL	(8,2)	NOT NULL	Baseline reference value
EffectiveFromUTC	TIMESTAMP	-	NOT NULL	Start of validity (UTC)
EffectiveToUTC	TIMESTAMP	-	NULL	End of validity (UTC)

Table A-10: Data dictionary for **BaselineProfile** entity

Entity: AnomalyAlert				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
AlertID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique anomaly alert identifier
MissionID	VARCHAR	36	FK → MissionRun.MissionID, NOT NULL	Mission where anomaly occurred
VehicleID	VARCHAR	20	FK → Vehicle.VehicleID, NOT NULL	Vehicle involved
AlertType	ENUM	-	'Power', 'Pressure', 'Temp', 'Comms'	Alert type
Deviation_pct	DECIMAL	(6,2)	NOT NULL	Deviation from baseline (percent)
Severity	ENUM	-	'INFO', 'WARNING', 'CRITICAL'	Severity level shown in UI
DetectedAtUTC	TIMESTAMP	-	NOT NULL	Detection time (UTC)
Status	ENUM	-	'Open', 'Acknowledged', 'Resolved'	Alert lifecycle state
ReviewedBy	VARCHAR	20	FK → Operator.OperatorID, NULL	Operator who reviewed/responded
ReviewedAtUTC	TIMESTAMP	-	NULL	Review time (UTC)

Table A-11: Data dictionary for **AnomalyAlert** entity

Entity: OperationalLogRecord				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
LogID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique log record
TimestampUTC	TIMESTAMP	-	NOT NULL	Timestamp (UTC) shown in data logs
MissionID	VARCHAR	36	FK → MissionRun.MissionID, NULL	Optional mission context
VehicleID	VARCHAR	20	FK → Vehicle.VehicleID, NOT NULL	Vehicle ID shown in data logs
OperatorID	VARCHAR	20	FK → Operator.OperatorID, NULL	Operator ID shown in data logs (if applicable)
EventType	VARCHAR	50	NOT NULL	Event type shown in data logs
Description	TEXT	-	NOT NULL	Event description shown in data logs
Severity	ENUM	-	'INFO', 'WARNING', 'CRITICAL'	Severity badge and filter

Table A-12: Data dictionary for **OperationalLogRecord** entity

Entity: ReportingJob				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
JobID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique job identifier
JobType	ENUM	-	'ExportFilteredLogs', 'PostMissionReport', 'ExportFullAuditLog'	Type of reporting/export action
RequestedBy	VARCHAR	20	FK → Operator.OperatorID, NOT NULL	Requesting operator
RequestedAtUTC	TIMESTAMP	-	NOT NULL	Request time (UTC)
CompletedAtUTC	TIMESTAMP	-	NULL	Completion time (UTC)
ExportFormat	ENUM	-	'CSV', 'JSON'	Export format selected in UI
FilterVehicleID	VARCHAR	20	FK → Vehicle.VehicleID, NULL	Vehicle filter (if used)
FilterSeverity	ENUM	-	'INFO', 'WARNING', 'CRITICAL', NULL	Severity filter (if used)
SearchText	VARCHAR	120	NULL	Search keyword applied to logs (if used)
OutputLocation	VARCHAR	255	NULL	Saved output path/link
Status	ENUM	-	'Queued', 'Running', 'Completed', 'Failed'	Job processing status

Table A-13: Data dictionary for **ReportingJob** entity

Entity: AuditLogRecord				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
AuditID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique audit event
TimestampUTC	TIMESTAMP	-	NOT NULL	Timestamp shown in audit log (UTC)
OperatorID	VARCHAR	20	FK $\rightarrow$ Operator.OperatorID, NULL	Operator who performed the action (if applicable)
Action	VARCHAR	80	NOT NULL	Action performed (e.g., Login, Mission deployed, Safe-State initiated)
Resource	VARCHAR	80	NULL	Target resource (e.g., ASV-03, Dashboard, Mission-12A)
Outcome	ENUM	-	'SUCCESS', 'FAILED'	Outcome of the action
Details	TEXT	-	NULL	Optional additional details

Table A-14: Data dictionary for **AuditLogRecord** entity

Entity: Certificate				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
CertID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique certificate record
CertName	VARCHAR	100	NOT NULL	Certificate name shown in UI
Issuer	VARCHAR	100	NOT NULL	Certificate issuer shown in UI
CertType	ENUM	-	'X.509'	Certificate type
LastRenewedAtUTC	TIMESTAMP	-	NULL	Last renewal time (UTC), if applicable
ExpiresAtUTC	TIMESTAMP	-	NOT NULL	Expiry timestamp used to compute remaining validity

Table A-15: Data dictionary for **Certificate** entity

Entity: SystemHealthCheck				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
HealthID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique health record
ServiceName	VARCHAR	60	NOT NULL	Service name shown in system health panel
Status	ENUM	-	’HEALTHY’, ’DEGRADED’, ’DOWN’	Health status indicator
Uptime_pct	DECIMAL	(5,2)	NOT NULL	Service uptime (percent)
LastCheckSecAgo	INTEGER	6	NOT NULL	Time since last check (seconds) shown in UI
CheckedAtUTC	TIMESTAMP	-	NOT NULL	Measurement time (UTC)

Table A-16: Data dictionary for **SystemHealthCheck** entity

Entity: SecurityConfiguration				
Attribute	**Data Type**	**Length**	**Constraints**	**Description**
ConfigID	VARCHAR	36	PK, NOT NULL, UNIQUE	Unique configuration record
SessionTimeoutMin	INTEGER	4	NOT NULL	Session timeout value (minutes) selected in UI
TwoFactorPolicy	ENUM	-	’RequiredForAdmins’, ’RequiredForAllUsers’, ’Optional’	Two-factor authentication policy
PasswordPolicy	ENUM	-	’Standard’, ’HighSecurity90Days’	Password policy selection
UpdatedBy	VARCHAR	20	FK → Operator.OperatorID, NOT NULL	Administrator who updated the settings
UpdatedAtUTC	TIMESTAMP	-	NOT NULL	Settings update time (UTC)

Table A-17: Data dictionary for **SecurityConfiguration** entity

Appendix B: Response Sequence

Use Case No.	Use Case Flow (Description)	Use Case End Goal
UC01 – Initiate Emergency Retrieval	**Main Flow:** 1. Pilot detects an emergency situation. 2. Pilot selects the emergency retrieval command. 3. System validates hardware interfaces. 4. System activates retrieval protocol. 5. Vessel begins return-to-surface procedure.	The vessel is safely retrieved during an emergency.
UC02 – Monitor Vessel Health	**Main Flow:** 1. System continuously receives vessel telemetry. 2. System checks pressure, temperature, and power levels. 3. System compares readings with safety thresholds. 4. System detects abnormal readings if thresholds are exceeded. 5. System prepares alerts for operators.	Vessel health is monitored continuously to detect issues early.
UC03 – Manual Control Override	**Main Flow:** 1. Operator selects the manual override option. 2. System verifies operator authorization. 3. System disables autonomous control temporarily. 4. Operator sends manual navigation commands.	Operator gains direct control of the vessel when necessary.
UC04 – Execute Mission Controls Override	**Main Flow:** 1. System receives manual control commands. 2. System validates command parameters. 3. System applies override to mission controls. 4. System logs operator actions. 5. Vessel executes the override instructions.	Mission continues under manual control safely.
UC05 – Configure Mission Parameters	**Main Flow:** 1. Fleet Controller opens mission configuration interface. 2. Controller enters mission parameters (depth, path, duration). 3. System validates parameter limits. 4. Controller confirms configuration. 5. System stores mission parameters.	Mission parameters are prepared for safe mission execution.
UC06 – Manage Fleet Mission	**Main Flow:** 1. Fleet Controller views fleet dashboard. 2. Controller assigns missions to vessels. 3. System distributes mission data to vessels. 4. Controller monitors fleet progress. 5. System updates fleet status in real time.	Multiple vessels are coordinated efficiently.
UC07 – Review Mission Log	**Main Flow:** 1. Mission Review Officer accesses mission log system. 2. Officer selects a completed mission. 3. System retrieves stored mission records. 4. System displays full mission timeline.	Mission performance is evaluated using stored logs.
UC08 – Synchronise Mission Data	**Main Flow:** 1. System collects mission data from vessels. 2. System encrypts data using secure protocols. 3. System synchronizes data with control station servers. 4. System verifies successful synchronization. 5. System updates mission data repository.	Mission data remains consistent across all systems.

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Use Case No.	Use Case Flow (Description)	Use Case End Goal
UC09 – Auto-Trigger Retrieval	**Main Flow:** 1. System detects abnormal operating conditions. 2. System evaluates emergency thresholds. 3. System automatically triggers retrieval protocol. 4. Vessel receives retrieval command. 5. Vessel begins safe recovery procedure.	Automatic safety retrieval occurs during critical failures.
UC10 – Detect Abnormal Conditions	**Main Flow:** 1. System continuously monitors sensor data. 2. System evaluates pressure, temperature, and power metrics. 3. System compares metrics to defined limits. 4. System identifies abnormal condition. 5. System notifies alert subsystem.	Abnormal system conditions are detected quickly.
UC11 – Raise Health Alert	**Main Flow:** 1. System receives abnormal condition signal. 2. System generates health alert notification. 3. Alert is displayed on operator dashboard. 4. System logs the alert event. 5. Operator reviews and decides next action.	Operators are notified of system health issues.
UC12 – Validate Hardware Interfaces	**Main Flow:** 1. System checks connected hardware modules. 2. System tests communication interfaces. 3. System verifies sensor availability. 4. System confirms interface compatibility. 5. System reports validation status.	Hardware readiness is verified before operations.
UC13 – Log Operator Operations	**Main Flow:** 1. Operator performs system action. 2. System captures action details. 3. System records timestamp and user identity. 4. System stores log entry securely. 5. Audit log is updated.	Operator activities are recorded for auditing.
UC14 – Autonomous Execution	**Main Flow:** 1. System loads configured mission parameters. 2. Vessel begins autonomous navigation. 3. System monitors execution progress. 4. System records telemetry data. 5. Mission continues until completion or interruption.	Vessel completes mission autonomously.
UC15 – Log Mission Data	**Main Flow:** 1. System collects mission telemetry data. 2. System records sensor readings. 3. System timestamps each data entry. 4. Data is formatted for storage. 5. Log records are prepared for database storage.	Mission telemetry data is recorded during operations.
UC16 – Store Mission Data	**Main Flow:** 1. System receives mission log records. 2. System encrypts stored data. 3. System writes records to database storage. 4. System confirms successful storage. 5. Data repository is updated.	Mission data is securely stored in the system database.

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Use Case No.	Use Case Flow (Description)	Use Case End Goal
UC17 – Retrieve Mission Data	**Main Flow:** 1. Officer requests mission data retrieval. 2. System searches mission database. 3. System retrieves requested data. 4. System formats data for display. 5. Data is presented to the officer.	Stored mission data becomes accessible for analysis.

Table B-1: Response Sequence: Use Case Flows

Appendix C: Response Sequence (Data Dictionary)

Data Dictionary Entity	Data Dictionary Flow (Description)	Data Dictionary End Goal
Operator	**Main Flow:** 1. Admin creates or updates an operator account (role, account status, 2FA). 2. System checks required fields and unique IDs/usernames. 3. System stores the operator record with created time. 4. System shows the operator in the admin user list. 5. Operator can be referenced by sessions, commands, logs, and audit records.	Operator identities exist to control access and attribute actions.
LoginSession	**Main Flow:** 1. Operator submits login credentials. 2. System verifies operator account and authorization. 3. If required, system verifies 2FA. 4. System creates a session record (login time, 2FA result, session state). 5. System updates the operator last-login time.	Login activity is tracked for access control and auditing.
Vehicle	**Main Flow:** 1. System registers vehicles using VehicleID and display name. 2. System receives telemetry readings (depth, power, pressure, temp, position). 3. System updates each vehicle's latest state for the live dashboard. 4. System stores last update time. 5. Operators view fleet status using the latest stored state.	Vehicles are identifiable and their latest live state is available for monitoring.
MissionPlan	**Main Flow:** 1. Fleet controller creates a mission plan (name, version). 2. Controller selects a target vehicle and enters mission settings (e.g., duration). 3. System validates inputs and stores the mission plan with creator details. 4. Mission plan is saved as Draft until deployment. 5. Mission plan can be deployed to create a mission run.	A validated mission plan is stored for later deployment and execution.
Waypoint	**Main Flow:** 1. Controller adds waypoints to a mission plan (lat, lon, depth, phase). 2. System assigns or updates waypoint order (sequence). 3. System validates waypoint values and stores them under the mission plan. 4. Waypoints can be reordered and saved again. 5. The mission plan holds an ordered route for the vehicle to follow.	An ordered mission route exists as waypoints linked to the mission plan.
MissionRun	**Main Flow:** 1. System deploys a mission plan and creates a mission run record. 2. System records mission start time and initial state. 3. System updates state during operation (running/paused/safe-state/completed/terminated). 4. System records mission end time when the run ends. 5. System stores mission KPI summary (e.g., waypoint success percent).	Mission execution is tracked as a run with states, timing, and outcome summary.

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Data Dictionary Entity	Data Dictionary Flow (Description)	Data Dictionary End Goal
Telemetry-Snapshot	**Main Flow:** 1. System receives telemetry during a mission run. 2. System timestamps each reading (UTC). 3. System stores depth, power, pressure, temp, and position per snapshot. 4. Each snapshot is linked to the mission run and vehicle. 5. Telemetry can be used for monitoring, alerts, and reporting.	Time-series telemetry is stored for traceability and analysis.
CommsLink-Snapshot	**Main Flow:** 1. System measures communications link metrics during operation. 2. System timestamps each link snapshot (UTC). 3. System stores link status, quality, bandwidth, and latency. 4. Each snapshot is linked to the mission run. 5. Operators view link health and degradation over time.	Communications health is tracked to support reliable operations.
Intervention-Command	**Main Flow:** 1. Operator selects a control action (pause/resume/safe-state). 2. System verifies operator authorization for the action. 3. System issues the command to the target vehicle. 4. System records delivery outcome (acknowledged/failed) with timestamps. 5. Command history supports review and accountability.	Manual control actions are recorded with attribution and outcomes.
Baseline-Profile	**Main Flow:** 1. System stores baseline values per vehicle and metric type. 2. Baselines have effective date range (from/to). 3. Monitoring logic uses the active baseline for comparisons. 4. Baselines can be updated by creating a new effective range. 5. Baselines support consistent deviation calculations.	Baseline references exist for detecting deviations from normal behaviour.
AnomalyAlert	**Main Flow:** 1. System detects abnormal readings based on rules/baselines. 2. System creates an alert with type, severity, deviation percent, and time. 3. Alert is shown to operators for review. 4. Operator can acknowledge or resolve the alert. 5. Review details (who/when) are stored for traceability.	Abnormal conditions are recorded and can be reviewed to support response.
Operational-LogRecord	**Main Flow:** 1. A mission event or operator action occurs. 2. System records event type and description with UTC timestamp. 3. System links the record to vehicle and (if applicable) mission/operator. 4. System stores severity level for filtering. 5. Logs can be searched, filtered, and exported.	Operational events are stored for investigation, reporting, and review.
ReportingJob	**Main Flow:** 1. Operator requests an export/report (type + format + filters). 2. System stores the job request and sets status to queued/running. 3. System generates output (e.g., CSV/JSON export or post-mission report). 4. System records completion time and output location. 5. Job status updates to completed/failed.	Exports and reports are generated and tracked with job history and outputs.

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Data Dictionary Entity	Data Dictionary Flow (Description)	Data Dictionary End Goal
AuditLogRecord	**Main Flow:** 1. A security-relevant action happens (login, deploy, safe-state, config change). 2. System records timestamp (UTC) and actor (if available). 3. System stores action, resource, and outcome (success/failed). 4. Audit records are retained for accountability. 5. Audit log can be exported if needed.	Security-sensitive activity is traceable through audit records.
Certificate	**Main Flow:** 1. System stores certificate identity and issuer details. 2. System records last renewed time when renewal occurs. 3. System stores expiry time for validity tracking. 4. UI can compute “expires in X days” from expiry time. 5. Certificate records support secure system communications.	Certificate validity is tracked so secure connections remain enforceable.
SystemHealth-Check	**Main Flow:** 1. System checks key services (database, telemetry, comms, authentication). 2. System sets status (healthy/degraded/down) with uptime percent. 3. System stores “seconds since last check” and measurement timestamp. 4. UI shows latest health status to operators. 5. Health history can support troubleshooting.	System services are monitored and stored for operational visibility.
Security-Configuration	**Main Flow:** 1. Admin selects session timeout, 2FA policy, and password policy. 2. System validates the settings. 3. System stores the configuration with updater and update time (UTC). 4. System enforces the updated policies for access and sessions. 5. Changes are traceable and can be audited.	Security policies are stored and enforced with clear change attribution.

Table C-1: Data Dictionary Response Sequence Flows